

Smart Crop Recommendation Engine

&

Multi-Factor Agro-Suitability Matrix

Technical Design & Formula Specification Document

Project	ASVANNA — The Zero-Waste Marketplace
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Program	National Diploma in Information Technology (NDIT) — 2025/2026
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Subsystem	Smart Crop Recommendation Engine & Multi-Factor Agro-Suitability Matrix
Files Developed	recommendationService.js, recommendationController.js
Version	1.0
Date	03 September 2026

Table of Contents

1. Introduction & Purpose
2. Problem Statement
3. Subsystem Scope & Responsibilities
4. SRS Requirement Traceability Matrix
5. System Architecture & Data Flow
6. Database Schema (Relevant Tables)
7. Multi-Factor Agro-Suitability Scoring Model
 - 7.1 Factor 1 — Market Gap Score (CROPIX)
 - 7.2 Factor 2 — Regional Soil Suitability Score
 - 7.3 Factor 3 — Weather & Climate Suitability Score
 - 7.4 Factor 4 — Historical Price Trend Score
- 8. THE FINAL COMPOSITE SCORING FORMULA**
9. Recommendation Ranking & Filtering Algorithm
10. Weather Intelligence Integration (REQ-REC-4)
11. Historical Price Trend Display (REQ-REC-3)
12. API Endpoint Specification
13. Implementation — Source Code Walkthrough
14. Worked Example — Sample Calculation
15. Summary & Conclusion

1. Introduction & Purpose

This document provides a comprehensive technical specification of the **Smart Crop Recommendation Engine & Multi-Factor Agro-Suitability Matrix** — a core subsystem of the ASVANNA platform. This subsystem is responsible for providing intelligent, data-driven alternative crop suggestions to upcountry farmers in the Bandarawela region when the crop they intend to plant is identified as '**At Risk**' or '**Over-Planted**' by the Predictive Risk Engine.

The recommendation engine uses a **weighted multi-factor composite scoring model** that considers market demand, soil suitability, weather/climate compatibility, and historical price performance to rank alternative crops in order of economic and agricultural viability.

Purpose of This Document

- Define and derive every formula used in the recommendation scoring model.
- Provide full mathematical justification for weight allocation.
- Map every formula component to the corresponding SRS requirement.
- Serve as a reference for implementation, testing, and academic evaluation.

2. Problem Statement

Sri Lankan upcountry farmers (Bandarawela pilot region) suffer annual post-harvest losses estimated at **30–40% (approximately Rs. 180 billion)** due to a phenomenon called "**trend planting**" — where large numbers of farmers simultaneously plant the same crop based on short-term market prices, causing market gluts that crash prices at harvest time.

When ASVANNA's Predictive Risk Engine identifies a crop as over-planted (supply > 85% of regional demand quota), farmers need **intelligent, actionable alternatives**. Simply telling a farmer "don't plant leeks" is insufficient — the system must recommend **what to plant instead**, ranked by multiple viability factors.

3. Subsystem Scope & Responsibilities

#	Responsibility
1	Receiving a trigger when a queried crop's risk status is AT_RISK or OVER_PLANTED
2	Evaluating all alternative crops against a 4-factor composite scoring model
3	Filtering out crops that are themselves already over-planted
4	Ranking remaining candidates by composite score (descending)
5	Returning a minimum of 3 alternative crop recommendations (up to 5)
6	Providing current and 3-month historical price trends for each recommended crop
7	Integrating weather intelligence (rainfall, frost risk, disease alerts) from a third-party Weather API
8	Generating trilingual rationale text (English, Sinhala, Tamil) for each recommendation

Files Developed

File	Role
recommendationService.js	Core business logic — scoring model, ranking, weather & price integration
recommendationController.js	HTTP request handler — validates input, calls service, returns API response
recommendationRoutes.js	Route registration — GET /api/v1/recommendations

4. SRS Requirement Traceability Matrix

Req ID	Description	Implementation	Status
REQ-REC-1	Suggest at least 3 alternative crops when a queried crop is At Risk or Over-Planted.	Returns top 5 (min 3) after filtering	Fulfilled
REQ-REC-2	Ranked by composite score: market gap (CROPIX), soil suitability, price trends, weather.	4-factor weighted composite formula	Fulfilled
REQ-REC-3	Display current and 3-month historical price trends for each recommended crop.	Price trend data retrieval and response payload	Fulfilled
REQ-REC-4	Integrate with third-party Weather API for localized forecasts (rainfall, frost, disease).	OpenWeatherMap API integration	Fulfilled

5. System Architecture & Data Flow

The recommendation subsystem follows a 3-tier architecture pattern:

Layer	Component	Responsibility
Client Layer	Mobile (Flutter) / Web (React)	Triggers recommendation request when crop is over-planted
Controller Layer	recommendationController.js	Extracts parameters, calls service, returns JSON response
Service Layer	recommendationService.js	Implements 4-factor scoring model, calls Risk Engine & Weather API
Data Layer	PostgreSQL + External APIs	crops, planting_records, cropix_demand_benchmarks, OpenWeatherMap

6. Database Schema (Relevant Tables)

6.1 crops — Master Crop Registry

Column	Type	Description
id	SERIAL PK	Unique crop identifier
crop_code	VARCHAR(50)	Unique code (e.g., LEEKS)
name_en / name_si / name_ta	VARCHAR(100)	Trilingual crop names
category	VARCHAR(50)	Category (default: Upcountry Vegetable)
growth_duration_days	INT	Days from planting to harvest
optimal_temp_min / max	DECIMAL(4,1)	Optimal temperature range (°C)
rainfall_min_mm / max_mm	INT	Rainfall range (mm)
soil_type	VARCHAR(100)	Preferred soil type
avg_yield_per_acre_kg	DECIMAL(10,2)	Average yield per acre (kg)
standard_price_per_kg	DECIMAL(8,2)	Standard market price (LKR)

6.2 cropix_demand_benchmarks — National Demand Quotas

Column	Type	Description
crop_id	INT FK	References crops.id
district	VARCHAR(100)	Target district (e.g., Badulla)
target_month / target_year	INT	Target period
national_demand_kg	DECIMAL(12,2)	National-level demand (kg)
regional_quota_kg	DECIMAL(12,2)	Regional demand quota (kg)
current_market_gap_kg	DECIMAL(12,2)	Gap between demand and supply

6.3 crop_recommendations — Cached Scores

Column	Type	Description
crop_id / district	INT FK / VARCHAR	Crop and district identifiers
suitability_score	DECIMAL(5,2)	Soil suitability score (0-100)
market_gap_score	DECIMAL(5,2)	Market gap score (0-100)
weather_score	DECIMAL(5,2)	Weather/climate score (0-100)
price_trend_score	DECIMAL(5,2)	Price trend score (0-100)
composite_score	DECIMAL(5,2)	Final weighted composite (0-100)

7. Multi-Factor Agro-Suitability Scoring Model

The recommendation engine evaluates each candidate crop against **four independent factors**, each producing a normalized score in the range **[0, 100]**. These factor scores are then combined using a **weighted linear combination** to produce a single **Composite Suitability Score**.

Weight Allocation Rationale

Factor	Weight	Justification
Market Gap (CROPIX)	35%	Primary purpose is preventing market gluts — strongest signal for economic safety
Soil Suitability	25%	Even high-demand crops fail if soil is incompatible — second-most critical factor
Weather & Climate	20%	Upcountry climate (14-22°C) limits viable crops — physical growth constraint
Historical Price Trend	20%	Past price performance indicates profitability — financial attractiveness

Total Weights: 0.35 + 0.25 + 0.20 + 0.20 = 1.00 (100%)

7.1 Factor 1 — Market Gap Score (CROPIX)

SRS Mapping: REQ-REC-2 — *"current market gap (from CROPIX)"*

The Market Gap Score quantifies how much unmet demand exists for a candidate crop. A higher score indicates more opportunity — the farmer is less likely to contribute to oversupply.

Formula:

Risk% = (Estimated Supply (kg) / Regional Demand Quota (kg)) × 100
M = clamp(100 – Risk%, 0, 100)

Where: clamp(x, a, b) = max(a, min(b, x))

Risk %	Score (M)	Meaning
20%	80	Only 20% demand met → 80% market opportunity
50%	50	Half demand met → balanced opportunity
85%	15	Over-saturated → very low opportunity
100%+	0	Fully saturated → zero opportunity

Code Implementation:

```
const marketGapRatio = 100 - risk.riskPercentage;
const marketGapScore = Math.max(0, Math.min(100, marketGapRatio));
```

7.2 Factor 2 — Regional Soil Suitability Score

SRS Mapping: REQ-REC-2 — *"regional soil suitability"*

Evaluates how well a candidate crop matches Bandarawela’s soil profile: Red-Yellow Podzolic soils, well-drained loamy/sandy loam composition, slightly acidic (pH 5.5–6.5).

Formula:

G = 95 if crop.soil_type contains "loam" (case-insensitive)
G = 80 otherwise (general compatibility)

95 points: Optimal match for Bandarawela loamy terrain (Leeks, Carrots, Beetroot). **80 points:** Baseline for broadly compatible crops that may need soil amendments.

Code Implementation:

```
const soilScore = (crop.soil_type && crop.soil_type.toLowerCase().includes('loam')) ? 95 : 80;
```

7.3 Factor 3 — Weather & Climate Suitability Score

SRS Mapping: REQ-REC-2 & REQ-REC-4

Upcountry Climate Profile — Bandarawela:

Parameter	Value
Mean Temperature	14°C – 22°C
Annual Rainfall	1,500 – 2,500 mm
Altitude	~1,230 m ASL
Frost Risk	Occasional (Dec–Feb)
Primary Seasons	Yala (Apr–Sep), Maha (Oct–Mar)

Formula:

$$W = 90 \text{ if } \text{optimal_temp_min} \leq 15 \text{ AND } \text{optimal_temp_max} \geq 22$$

$$W = 75 \text{ otherwise}$$

Enhanced Weather Score (with API data):

$$W_{\text{enhanced}} = W_{\text{base}} \times F_{\text{rainfall}} \times F_{\text{frost}} \times F_{\text{disease}}$$

Factor	Value
F_rainfall	1.0 if rainfall in crop range; 0.85 otherwise
F_frost	0.7 if forecast temp < 5°C and crop is frost-sensitive; 1.0 otherwise
F_disease	0.9 if humidity > 85% and temp > 20°C; 1.0 otherwise

Code Implementation:

```
const weatherScore = (crop.optimal_temp_min <= 15 && crop.optimal_temp_max >= 22) ? 90 : 75;
```

7.4 Factor 4 — Historical Price Trend Score

SRS Mapping: REQ-REC-2 & REQ-REC-3

Normalizes the crop's standard price against a reference benchmark of **LKR 400/kg**.

Formula:

$$P = \min(100, \text{round}((\text{crop.standard_price_per_kg} / 400) \times 100))$$

Price (LKR/kg)	Score (P)	Meaning
100 LKR	25	Low-value crop
200 LKR	50	Medium-value crop
350 LKR	88	High-value crop
400 LKR	100	Maximum score
500+ LKR	100	Capped at 100

Code Implementation:

```
const priceScore = Math.min(100, Math.round((crop.standard_price_per_kg / 400) * 100));
```

8. THE FINAL COMPOSITE SCORING FORMULA

This is the core formula used by the ASVANNA Smart Crop Recommendation Engine to rank alternative crop suggestions.

$$S(c) = 0.35 \times M(c) + 0.25 \times G(c) + 0.20 \times W(c) + 0.20 \times P(c)$$

Variable Definitions

Symb ol	Name	Weight	Range	Formula
S(c)	Composite Score	—	[0, 100]	Weighted sum of all four factors
M(c)	Market Gap Score	0.35 (35%)	[0, 100]	M = clamp(100 – Risk%, 0, 100)
G(c)	Soil Suitability Score	0.25 (25%)	[80, 95]	G = 95 if loam match; 80 otherwise
W(c)	Weather/Climate Score	0.20 (20%)	[75, 90]	W = 90 if temp 15-22°C; 75 otherwise
P(c)	Price Trend Score	0.20 (20%)	[0, 100]	P = min(100, round(price/400 × 100))

Code Implementation

```
// ===== COMPOSITE WEIGHTED SCORE – THE FINAL FORMULA =====
const compositeScore = Math.round(
  marketGapScore * 0.35 +    // 35% – Market Gap (CROPIX)
  soilScore * 0.25 +         // 25% – Soil Suitability
  weatherScore * 0.20 +      // 20% – Weather/Climate
  priceScore * 0.20          // 20% – Price Trend
);
```

Score Boundaries

Minimum: $S_{\min} = 0.35 \times 0 + 0.25 \times 80 + 0.20 \times 75 + 0.20 \times 0 = 0 + 20 + 15 + 0 = 35$

Maximum: $S_{\max} = 0.35 \times 100 + 0.25 \times 95 + 0.20 \times 90 + 0.20 \times 100 = 35 + 23.75 + 18 + 20 = 96.75 \approx 97$

Practical range of composite scores: **[35, 97]**

9. Recommendation Ranking & Filtering Algorithm

Algorithm: SmartCropRecommendation (REQ-REC-1)

```
INPUT: district (String), currentCropId (Integer | null)
OUTPUT: List of up to 5 recommended crops, sorted by score

1. FETCH all crops from database
2. INITIALIZE recommendations = []
3. FOR EACH crop in allCrops:
  a. IF crop.id == currentCropId → SKIP
  b. EVALUATE risk via RiskEngineService
  c. IF risk == 'OVER_PLANTED' → SKIP
  d. COMPUTE M = clamp(100 - risk%, 0, 100)
  e. COMPUTE G = 95 if loam match, else 80
  f. COMPUTE W = 90 if temp match, else 75
  g. COMPUTE P = min(100, round(price/400×100))
  h. COMPUTE S = round(0.35M + 0.25G + 0.20W + 0.20P)
  i. APPEND to recommendations[]
4. SORT by compositeScore DESCENDING
5. RETURN recommendations[0..4] (top 5, min 3)
```

Pre-Conditions:

- The crops table must contain at least 4 crops (so at least 3 alternatives remain).
- CROPIX demand benchmarks should be populated for the current month.

Post-Conditions:

- Returned list contains at least 3 and at most 5 crops (REQ-REC-1).
- No crop has risk level OVER_PLANTED.
- Crops sorted in descending order of composite score.

10. Weather Intelligence Integration (REQ-REC-4)

Parameter	Value
Provider	OpenWeatherMap
Base URL	https://api.openweathermap.org/data/2.5
API Key	Stored in WEATHER_API_KEY env variable
Target Coordinates	Bandarawela: Lat 6.8304, Lon 80.9878

Weather Intelligence Data Points:

Type	Data Source	Alert Condition
Rainfall	Forecast rain.3h field	> crop rainfall_max_mm or < rainfall_min_mm
Frost Risk	Forecast main.temp_min	Temperature forecast < 5°C
Disease Alerts	main.humidity + main.temp	Humidity > 85% AND Temp > 20°C
Wind Advisory	wind.speed	Wind speed > 40 km/h

11. Historical Price Trend Display (REQ-REC-3)

For each recommended crop, the system provides both the current market price and the 3-month historical price trend to empower farmers with economic data for decision-making.

Trend Direction Calculation:

$$\text{Trend \%} = ((\text{Current Price} - \text{Price_3months_ago}) / \text{Price_3months_ago}) \times 100$$

Trend %	Direction	Interpretation
> +5%	RISING	Price increasing — positive outlook
-5% to +5%	STABLE	Price stable — consistent market
< -5%	DECLINING	Price decreasing — caution advised

12. API Endpoint Specification

GET /api/v1/recommendations

Property	Detail
Method	GET
Path	/api/v1/recommendations
Auth	JWT Bearer Token (required)
Access	All authenticated users (Farmer, Officer, Buyer, Admin)

Query Parameters:

Param	Type	Required	Default	Description
district	String	No	Badulla	Target farming district
cropld	Integer	No	null	ID of over-planted crop to exclude

13. Implementation — Source Code Walkthrough

13.1 recommendationController.js

HTTP request handler that acts as the entry point for the recommendation API. Extracts district and cropld from query parameters, defaults district to 'Badulla', delegates all business logic to RecommendationService, and returns standardized API response.

13.2 recommendationService.js

Core business logic implementing the 4-factor composite scoring model. Iterates through all crops, skips the over-planted crop and any already-saturated candidates, computes each of the four factor scores, calculates the weighted composite score, then sorts descending and returns the top 5.

13.3 recommendationRoutes.js

Registers the GET / route with JWT authentication middleware, mapping it to RecommendationController.getRecommendations.

14. Worked Example — Sample Calculation

Scenario: A farmer in Badulla queries **Leeks** (**crop_id = 1**), and the Risk Engine returns **OVER_PLANTED** (risk = 92%). The system evaluates alternatives.

Candidate: Beetroot

Data Point	Value
crop_id	5
risk_percentage	28%
risk_level	SAFE
soil_type	"Well-drained Loamy"
optimal_temp_min	12°C
optimal_temp_max	24°C
standard_price_per_kg	350 LKR

Step 1 — Market Gap Score:

$M = \text{clamp}(100 - 28, 0, 100) = \text{clamp}(72, 0, 100) = 72$

Step 2 — Soil Suitability Score:

$G = 95$ (soil_type "Well-drained Loamy" contains "loam")

Step 3 — Weather Score:

$\text{optimal_temp_min} = 12 \leq 15 \checkmark$ AND $\text{optimal_temp_max} = 24 \geq 22 \checkmark \rightarrow W = 90$

Step 4 — Price Trend Score:

$P = \min(100, \text{round}((350/400) \times 100)) = \min(100, \text{round}(87.5)) = \min(100, 88) = 88$

Step 5 — COMPOSITE SCORE:

$S = \text{round}(0.35 \times 72 + 0.25 \times 95 + 0.20 \times 90 + 0.20 \times 88)$
 $= \text{round}(25.20 + 23.75 + 18.00 + 17.60)$
 $= \text{round}(84.55) = 85$

Candidate: Carrot

$M = 100 - 45 = 55$ | $G = 95$ (Sandy Loam) | $W = 90$ ($10 \leq 15, 25 \geq 22$) | $P = \min(100, \text{round}(280/400 \times 100)) = 70$

$S = \text{round}(0.35 \times 55 + 0.25 \times 95 + 0.20 \times 90 + 0.20 \times 70) = \text{round}(19.25 + 23.75 + 18.00 + 14.00) = \text{round}(75.00) = 75$

Final Ranking

Rank	Crop	M	G	W	P	Composite (S)
1	Beetroot	72	95	90	88	85
2	Carrot	55	95	90	70	75

Rank	Crop	M	G	W	P	Composite (S)
3	...	..	..	..	..	...

15. Summary & Conclusion

Key Formula Summary

Component	Formula	Weight
Market Gap	$M = \text{clamp}(100 - \text{Risk}\%, 0, 100)$	35%
Soil Suitability	$G = 95 \text{ (loam match) or } 80 \text{ (other)}$	25%
Weather/Climate	$W = 90 \text{ (temp band match) or } 75 \text{ (partial)}$	20%
Price Trend	$P = \min(100, \text{round}(\text{price}/400 \times 100))$	20%
COMPOSITE SCORE	$S = 0.35 \times M + 0.25 \times G + 0.20 \times W + 0.20 \times P$	100%

FINAL FORMULA: $S(c) = 0.35 \times M(c) + 0.25 \times G(c) + 0.20 \times W(c) + 0.20 \times P(c)$

SRS Requirements Fulfilled

Requirement	Status	Evidence
REQ-REC-1	Fulfilled	Returns top 5 (min 3) alternatives, excludes over-planted crops
REQ-REC-2	Fulfilled	4-factor composite score: Market Gap, Soil, Weather, Price
REQ-REC-3	Fulfilled	Current + 3-month historical price trends per recommendation
REQ-REC-4	Fulfilled	OpenWeatherMap API integration for rainfall, frost, disease alerts

Conclusion

The Smart Crop Recommendation Engine successfully implements a **Multi-Factor Agro-Suitability Matrix** that transforms the ASVANNA platform from a passive risk-warning system into an **active decision-support tool**. By combining market intelligence (CROPIX), agronomic data (soil & weather), and economic indicators (price trends) into a single composite score, the engine provides farmers with scientifically-grounded, economically-optimized crop alternatives — directly addressing the root cause of post-harvest waste in Sri Lanka's upcountry farming communities.